

Assignment 4

1. What are the services provided by the data link layer to the network layer?
2. Explain one-bit sliding window protocol. Also, discuss the two scenarios that can occur in a sliding window.
3. Explain the piggybacking technique. State the advantages and disadvantages of piggybacking techniques.
4. Explain the sliding window protocol with the go-back-n and selective repeat strategies?
5. Explain the purpose of framing. Define framing. Explain all methods of Framing.
6. Explain the sender side of stop and wait protocol.
7. Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $x^3 + x + 1$. Suppose the message 11000 is to be transmitted. What is the value of the checkbit sequence $C_2C_1C_0$?
8. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. Find the message that should be transmitted.
9. Consider the data unit to be transmitted is-
10011001111000100010010010000100
Consider 8 bit checksum is used. Please find the data that should be transmitted.
10. Suppose we have a 4-bit message that we want to transmit: 111011. Now we apply the above steps in this hamming code example. Calculate the parity bits.